

Spectravideo

With software standardisation currently being a moot point in the computer industry, the implementation of the much discussed MSX system on the Spectravideo home computer could change the face of things to come. Tony Hetherington looks over the machine and pontificates on its worldwide repercussions.



The Spectravideo home computer is the first MSX machine to reach these shores. It represents a truly international venture since it was designed at the Spectravideo head office in New York, uses MSX Basic written by Microsoft in Japan and is built by Spectravideo in Hong Kong.

There are two different models available: the SV-318 which features a built-in joystick and the more expensive

SV-328 which offers 80k of RAM and a full-size keyboard. In their simplest form they are powerful home computers offering superb sound and graphics facilities. However, the machines can be expanded into disk systems using the industry standard CP/M operating system.

Hardware

The Spectravideo computer is housed in

a white, plastic case which measures 39.4cm x 22.1cm and slopes from a height of 7.6cm at the back to 4.3cm at the front.

The top of the machine is dominated by the keyboard which fills about two thirds of the space with the rest consisting of a raised section housing a cartridge slot. Along the rear of the case is the video output port, the cassette port and an expansion port designed to

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accept the Super Expander (more about this later). The nine volt power input is on the right hand side along with an on/off switch and two joystick ports.

The Spectravideo is connected to a standard television set by a generous 1.5m cable travelling via an external TV modulator to a video switch box. The power is supplied by a heavy duty unit which supplies nine volts at 1.5 amps, which is sufficient to drive the computer, a cassette recorder and one other peripheral.

The SV-318's keyboard consists of 71 rubber keys and a built-in joystick. The keys are well spaced out which makes them easy to use and a great improvement on the other rubber-keyed computers currently on the market. This ease of use is enhanced by the inclusion of a space bar and two shift keys.

In addition to the standard keys found on a machine in this price bracket are five function keys, three program control keys and six miscellaneous keys.

Each of the five function keys can execute one of two functions depending on whether or not the Shift key is pressed. The functions to be executed are displayed at the bottom of the screen display. All are user-definable but the keys are preset to functions such as RUN, LIST and CLOAD.

The three program control keys are: ENTER; STOP to halt a program; and the CONTROL key which facilitates the CTRL commands (for example, CTRL-S).

Finally, the other keys are: CAPS LOCK, complete with a red LED, CLS/HOME, which clears the screen and returns the cursor to the home position; INSERT and DELETE for program editing; and LEFT and RIGHT GRAPHIC. These last two keys are used to select either of the two graphics characters printed above each of the alphabetic keys.

To the right of the keys is the built-in joystick. This is only a four direction stick and is used mainly for editing. The stick itself is inserted into a grey circular pad which can be used more easily without the stick in place.

The SV-328 differs from its cheaper brother in its amount of memory and in the quality of its keyboard. It boasts a full-size, 86-key, typewriter-style keyboard with the SV-318's joystick being replaced by a numeric keypad

complete with arithmetic and cursor keys.

Early sales literature from Spectravideo claimed that the SV-328 had a built-in word processor and HELP programs. These were, however, discarded in favour of full MSX status but have left their mark in some now useless keys. The SV-328 has an unresponsive PRINT key and both machines have a select key which has no apparent use.

The size of the top of the keys on the SV-328 is the same as SV-318's rubber keys, but since the plastic keys taper out, the keyboard of the SV-328 seems to be rather cramped. Also there is no room to show the available graphics characters.

The inside of the machines is definitely a 'no go' area since it is encased in a

metal box, held in position by several screws which are in turn sealed by dollops of red plastic.

Cassette player

The cassette player is a dedicated recorder which connects to the back of the computer via a cable which houses power, audio, digital and motor control lines.

The player itself is a rather ugly looking machine. Its straight front and angled sides give it a distorted appearance and the dull brown keys and cassette cover do nothing to improve this image. However, this doesn't stop it being a very good cassette player which is just as well since it's the only recorder you can use with the Spectravideo.

As it's a dedicated recorder it doesn't



The SV-328 boasts a full-size, 86-key, typewriter-style keyboard



The SV-318 keyboard consists of 71 rubber keys and a built-in joystick

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require any tone or volume controls and therefore avoids the loading problems which plague other home micros. It is under full computer control which is ideal for tutorial style programs as is the facility of a parallel audio track. This can be recorded using the built-in microphone, and play back relevant instructions as a program is loading in.

I generally don't like dedicated cassette players since they add to the expense of buying a micro, but this is quite a good one, and, as I said earlier, it's the only game in town.

Super Expander

Should you wish to expand your Spectravideo computer then you will undoubtedly need the Super Expander. The Expander can be used with both the SV-318 and the SV-328 and it plugs into the large expansion slot at the rear of the machine.

It's slightly larger than the Spectravideo itself but sits neatly behind the computer. Constructed out of the same sturdy, white plastic and consisting of two symmetrical halves, the top half is easily removed to allow access to the insides. It has its own power supply — sufficient to drive up to seven devices — which occupies the right-hand third of the internal space.

Next to the power assembly are

seven expansion slots. These are numbered 0 to 6, with slot 6 being reserved for a floppy disk controller.

To the left of these slots is an area designed to store the TV modulator and any excess aerial lead. The lead is fed in through a hole in the front and out via a slot in the back panel. This is particularly convenient since the top of the Expander is strong enough to support a 15in television.

There are seven other slots cut out of the back panel for any other cables or wires from cards plugged into the expansion slots. Corresponding holes in the front panel are created by removing clipped on plastic panels which can be replaced should the cartridges be rearranged. The front of the cartridge is visible through this opening for ease of use.

Expansion cartridges

Four expansion cartridges were available for review: the floppy disk controller, 16k and 64k memory expansions and a Centronics interface.

Each of the cartridges is enclosed in a standard dark grey case, the bottom of which has a section removed so that the cartridge can be plugged into an expansion slot. The top and front panels have an identifying label. The front panel has an additional LED to

show that all is well. The back of the case is open which spoils its appearance since the enclosed electronic card extends out of the back of the case. This, however, allows easy access to any ports or interfaces for connections to peripherals.

The floppy disk controller has two of these ports since it can support two floppy disk drives and the Centronics interface has a printer port.

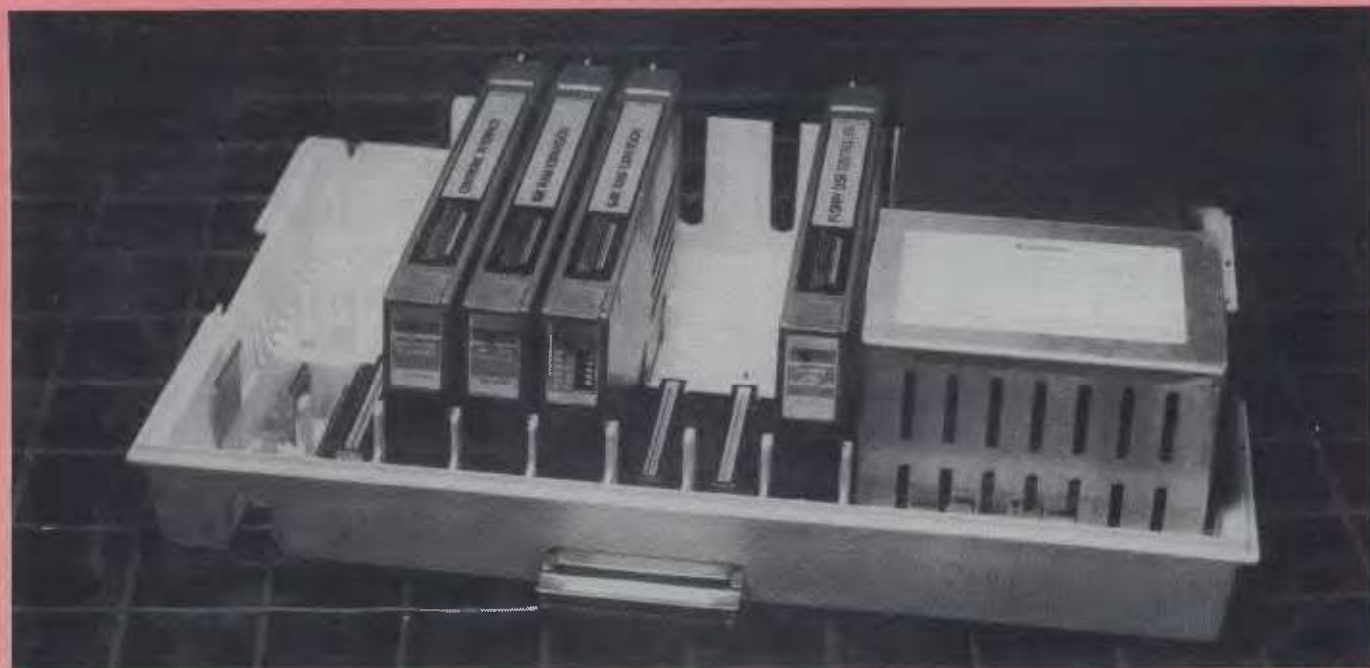
The 16k and 64k RAM cartridges can be used to extend the computer's memory up to 128k. This is beyond the addressable memory supported by the Z80 processor and therefore a system of banked switching must be employed. This is done manually by the setting of dip switches on the front of the 64k cartridges.

Floppy disk drive

The hardware available for review is completed by the inclusion of the floppy disk drive. This plugs into the floppy disk controller via a ribbon cable through which it draws its power and transmits and receives data.

The disk drive uses standard 5¼in single-sided, double-density disks and has a formatted capacity of about 164k.

The drive is quite noisy but this is excusable in a machine in this price bracket.



The Super Expander can be used with both the SV-318 and SV-328 and plugs in at the rear of the machine

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Up to two drives can be included to form a powerful CP/M-based system capable of many serious applications.

The expandability of the Spectravideo computer extends beyond the hardware supplied for review. Other products include a games adaptor to run Coleco games, a dot-matrix printer, a graphics tablet, an RS232 interface, an 80-column card and a modem.

System Software

When you switch on the Spectravideo the logo is displayed in three colours as the machine performs a diagnostic check. Once completed the screen reverts to its standard display of white text on a light blue background and reports the number of memory bytes free.

The SV-318 has 32k of RAM, although 16k of this is used for the graphics display. A further 3k is used with other system overheads leaving just under 13k for the user.

The SV-328 has just over 29k of user memory displayed but more is actually available. This is because the SV-328 has 80k which is beyond the addressable capacity of the Z80 processor. Therefore memory bank switching has been used so only 48k is in the machine at any time.

MSX Basic written by Microsoft is the operating language used and the facilities described here will be present in any MSX machine.

Writing your own programs is easy with MSX since it includes some features to ease the programmer's burden. AUTO automatically generates line numbers and RENUM rennumbers a finished program. The Spectravideo features a full screen editor which dramatically speeds up editing and correction of errors. Simply position the cursor over the error using the cursor keys or the joystick and correct the mistake.

The preset functions are set to aid the programmer and include RUN and LIST options, even LIST — to list the last line entered — and a colour function to reset the colours to their original setting.

MSX Basic includes most of the usual program structures but there are one or two curious omissions. The oddest of all is the exclusion of the WHILE-WEND structure since the Spectravideo's manual includes two error messages for this structure.

The error messages, should they occur in your programs, are typed out in full on the screen which is a great improvement on Microsoft Basic.

MSX includes a number of very useful interrupt control commands which can be used to great effect in programs. The most common of these is the ON ERROR GOSUB command which defines the subroutine to be executed should an error be detected. The others have a similar format and include ON INTERVAL which defines a routine which will be executed at set time intervals. ON KEY directs control to one of several destinations depending on which function key is pressed. ON STOP executes a routine when a program is halted by a user and ON SPRITE can be used to resolve collisions between sprites (for example, objects in a game). Finally ON STRIG calls a routine depending on the direction of a joystick.

Each of these interrupt commands can be turned on or off or reset after it has been triggered. Triggering these commands is automatic once they have been initiated and requires no further Basic commands. This will obviously speed up program execution since no Basic statements need be interpreted.

Arithmetically, the Spectravideo is extremely accurate as variables are automatically set to double precision. The price for this accuracy is loss of speed.

The Spectravideo supports a superb range of graphics commands. A single command performs a whole series of operations. For example, the command LINE(10,10)-(100,30),bf draws a rectangle between the two points specified and then fills it in. The coordinates in the example refer to the high resolution graphics screen which consists of 192 x 256 pixels. Each of these pixels can be PSET (set) or PRESET (reset) in any of the sixteen colours.

When the computer is switched on the screen resolution is in low resolution mode (40 x 24) in which only the block graphics characters can be selected using the left and right graphics keys. The SCREEN command is used to switch between screen modes.

Other advanced graphics facilities which you would normally expect to find in an extended Basic cartridge include: the CIRCLE command which

can draw circles and ellipses; PAINT which fills in a graphics shape in a specified colour; GET which stores a specified area of the screen in an array for printing anywhere on the screen using PUT.

The Spectravideo also supports up to 32 sprites at any one time. These sprites can be of two sizes: 8 x 8 or 16 x 16 pixels. The size to be used is selected as a field in the SCREEN statement.

Creation of sprites on the Spectravideo is a lot easier than on other machines, such as the Commodore 64 where details of the sprites' design must be entered into the computer using a series of POKES. To define a sprite on the Spectravideo all you need do is use the SPRITE command which assigns a sprite with the pattern held in a string of characters that can be easily created through READ and DATA statements.

The Spectravideo also possesses two Macro languages: a graphics macro language (GML); and a music macro language (MML). These are separate mini-languages existing with Basic.

The GML can be regarded as a graphics language similar in nature to Logo's turtle graphics. It consists of a series of single letter commands which are incorporated into the Basic command DRAW. These commands are directions given to a cursor that draws a line on the screen. Each command is followed by a number to complete the instruction. For example, U10 tells the cursor to go up 10 pixels. Other commands include: E,F,G and H to move diagonally; B, to move without leaving a line; A to set an angle; and C to set the colour of line drawn.

MML is similar to GML and provides the Spectravideo's sound capabilities. Quite simply the sequence of notes you wish to play is stored in a string and performed using the PLAY command. The command PLAY "CEG" would play the notes C, E and G. These basic notes can be tuned and modified by additional commands which include O to set the octave, T and L to determine the tempo and length of a note and S and M to control the shape and tone.

Up to three strings can be included in a PLAY command, separated by commas to set the notes to be played by each of the three sound channels.

The sound is generated by a separate chip which means that music can be

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produced without slowing down execution time.

This separate sound processor probably explains the difficulties I had in tuning in a television to use with the machine. I found it almost impossible to tune it so that the picture was sharp and the sound still audible. Eventually I was able to reach a balance between sound and vision and had no further problems.

Supplied with the floppy disk controller are two disk-based operating systems, MSX disk Basic and CP/M 2.2.

To be honest I really can't see the point of this version of MSX Basic. It contains only a few additions to the ROM-based version for accessing disks. Besides, the Spectravideo also supports a modified version of CP/M 2.2 which will be more than adequate for the needs of most users.

It can be implemented on a reasonably priced system with the minimum requirement being one floppy disk. To cater for this the CP/M utilities have been rewritten with a 40-column screen display. Additional utilities have been added for the user with only one disk drive including 1COPY which will copy a disk using only one drive. A whole disk is copied in four passes with clear instructions telling when to swap the disks.

Other Software

Commercial software is available for the Spectravideo in three different mediums: tape, cartridge and disk.

The cartridges are enclosed in dark grey cases which extend past the connecting port on three of the four sides. This guides the cartridge into the slot on the top of the machine. The cartridges fit firmly into their slots without being difficult to use by a child.

The cassette tapes are of high quality although any computer tape can be used to save your own programs.

Both the cassette and cartridge programs are supplied in attractive boxes complete with full instructions.

Four programs are included in the price of the machine: two games, or rather entertainment programs, entitled Spectron and Armour Assault, an "Introduction to Basic" course and a "Home Economist" package. Other games are also available and a couple are worth mentioning:

Crossforce is a blast-the-aliens type of game and is a mirrored version of Galaxians. Mirrored, because you control two laser bases, one at the bottom and the other at the top of the screen. The object is to catch the attacking aliens in crossfire between the two laser bases. There are three different ways of controlling the bases, parallel where they move together, diagonal where they move in opposite directions and dual control in which two players can play—each controlling a base.

Armour Assault is a variation of the old tank battle game but it has been brought up to date to form an enjoyable two player game. Eight different variations of the game exist, each different from the others in fuel, ammunition and tanks available to the players. Some options include hospitals to repair tanks and depots where you can refuel and rearm. Others include ghost tanks that try and destroy you.

Frantic Freddy is a rather curious game where you control the fearless fireman Freddy as he tries to put out a blazing building. According to the cover of the instruction book our hero Freddy looks like a hotdog sausage wearing boots, and with a funnel for a

nose. Through this funnel Freddy squirts water to extinguish a normal orange window fire but six to put out a purple fire that chases and tries to corner him. Freddy must also dodge flame particles or he will lose a life. The same fate awaits him should he fail to catch a cat as it tries to jump to safety.

The games are controlled by joysticks, either plugged into joystick ports or built into the SV-318. The cursor keys of the SV-328 can be used but this makes the games even harder than they already are.

The tutorial tape is billed as an introduction to programming in Basic and consists of a tape and a booklet. The idea is that the user can read through the booklet while working through examples on the tape. The examples given are interesting as is the explanatory text; however, it is dogged with errors. These are particularly bad since the package is designed for the beginner who will be confused by the mistakes. For example, having explained the difference between the Commands SCREEN 1 and SCREEN 2 the text refers to SCREEN + which is obviously wrong to all but a beginner.

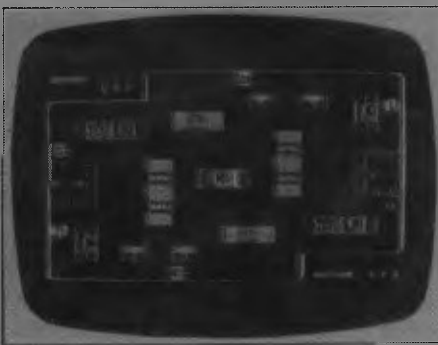
Benchmarks

BM1	2.3
BM2	5.7
BM3	18.2
BM4	20.0
BM5	20.9
BM6	32.8
BM7	45.1
BM8	235.7

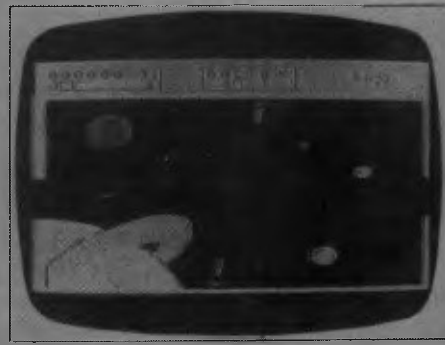
All timings in seconds. For a listing of the Benchmark programs see 'Direct Access'



Crossforce



Armour Assault



Frantic Freddy

The Perfect series of accounting software is available for the Spectravideo running under CP/M. The entire range is priced at \$799 but individual packages can be purchased. There is also a range of games etc. from Coleco which can be used on the Spectravideo with a Coleco adaptor priced at \$119.

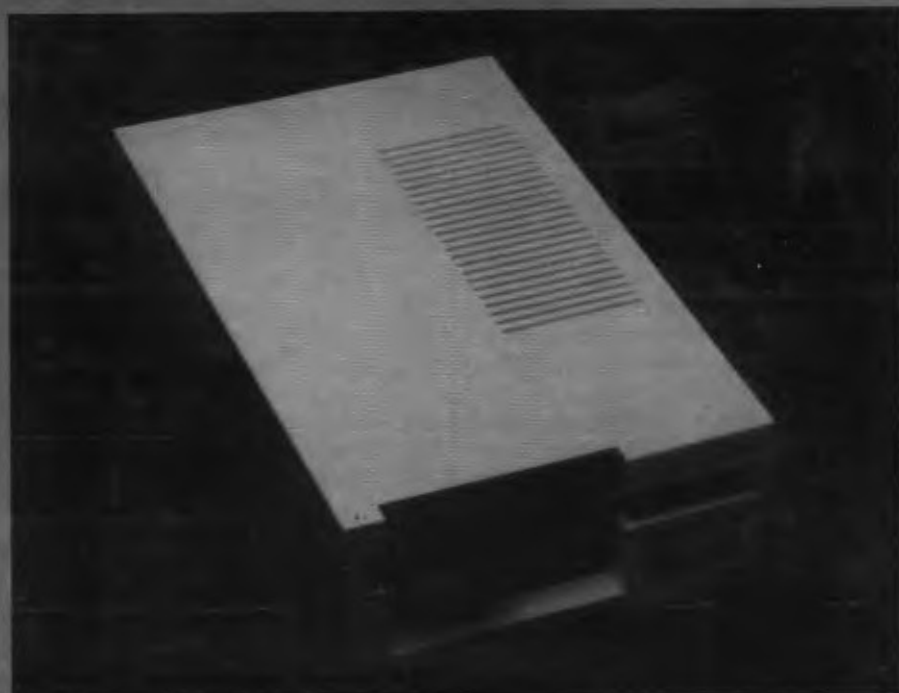
Documentation

At first sight the documentation for the Spectravideo seems plentiful and adequate. In fact there seems to be too much of it — even the RAM packs come complete with their own user manuals.

The manual which accompanies the computer itself is a 3-holed ring binder containing a loose leaf manual and a quick reference booklet. It is pleasantly organised, complete with separator cards between chapters which allow easy access to the information. Unfortunately the manual wants to keep the machine's secrets safe from the prying user which it does by ploys such as example programs that don't work, directions to tables that don't exist and, as mentioned earlier, error messages for program structures that aren't supported.

Small chapters may make the manual easy to read but they are inadequate for teaching Basic. Eighteen pages is hardly enough to cover all of the Spectravideo's excellent sound and graphic capabilities.

By far the best manual included in the review package was the CP/M manual



The disk drive uses standard 5 1/4 in single-sided, double density disks

which was mostly written by Digital Research. The section referring to features converted or specially written for the Spectravideo is concise and easy to understand. The more advanced features of CP/M will no doubt remain a mystery to all but the most determined user since the relevant sections of the manuals are daunting to the beginner.

Conclusions

Since the Spectravideo is the first MSX computer to reach Australia, my assessment of the machine is at two levels: as a machine in its own right and as the pioneering MSX micro.

The Spectravideo is priced to be in direct competition with the Commodore 64. It costs too much to be a serious

MSX — What is it?

The MSX system was proposed by Microsoft in Japan in an attempt to bring the standardisation of the record industry to the computer market. Such a system of standardisation is being welcomed by manufacturers, software houses and shops since they will all benefit from it.

Manufacturers of hardware, particularly of add on peripherals, will be able to produce goods for more than one machine at a time resulting in massive savings on research and production costs. Similar savings will be enjoyed by software houses with the additional benefit of a longer shelf life for their programs, since it is less likely that the machine for which they write will be superseded.

Invariably shops have serious shelf space problems since each computer needs a separate software display. MSX will solve this by requiring only one display for all machines.

MSX plans to achieve this standardisation by having at the centre of all computers Microsoft Extended Basic

running in a machine using a Z80 processor and featuring 16 colours and three sound channels. Since they will all have this standard format the software they use will be available for other MSX machines.

There are at present fifteen companies supporting MSX with Spectravideo being the only non-Japanese member. A full list of these is given below: it reads like a directory of Japanese electrical companies.

MSX is open to any company which wishes to join the scheme and is prepared to pay the admission fee. This may become an attractive package to companies thinking about entering the computer market since the computer design, particularly Basic, has been done for them.

This may be good for such companies but is it good for the computer market? Such standardising may seriously restrict developments and stagnate the industry as a consequence. Microsoft claims, however, that this will not happen since MSX is a very loose system allowing room for development. But the question is: Will

these developments be compatible with other MSX machines? I think not. I have seen literature written in Japanese claiming that the Sanyo MSX machine will be able to grab and store television pictures but the software required to run this probably won't work on a Spectravideo.

One final thought on MSX is: If there are many machines using the same software then software copying might become an even bigger problem than it is already. This is because copying will be possible where it is now impossible due to machine incompatibility.

Companies supporting MSX

Canon	NEC
Fujitsu	Sony
General	Pioneer
Hitachi	Sanyo
JVC	Spectravideo
Kyocera	Toshiba
Matsushita	Yamaha
Mitsubishi	